Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/25/2013
PI Name:	Limoli, Charles Ph.D.		
Project Title:	Oxidative stress as a mechanism for altering acute and chronic functional changes in the CNS exposed to low dose, low dose rates, and mixed fields of protons and HZE nuclei		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	climoli@uci.edu	**Fax:**	FY (949) 824-3566
PI Organization Type:	UNIVERSITY	**Phone:**	(949) 824-3053
Organization Name:	University of California		
PI Address 1:	Dept. of Radiation Oncology		
PI Address 2:	Medical Sciences I, B149		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	07/01/2009	**End Date:**	06/30/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: New end date is 6/30/2013 per A. Chu/ARC and NSSC information (Ed., 6/15/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Huang, Ting-Ting (VA Palo Alto Health Care System) Elmore, Eugene (University of California Irvine)		
Grant/Contract No.:	NNX09AK25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space radiation environment poses unique hazards that include solar particle events and galactic cosmic rays. These types of radiations have been shown to cause acute and long-lasting chemical changes referred to as oxidative stress that can alter the normal physiology of the central nervous system (CNS). The chemical species produced can cause a wide range of effects in cells within the CNS that largely depend on the types of pathways they activate and the types of damage they produce. Proper functioning of the CNS depends on many factors, but is known to require subpopulations of neural stem and precursor cells that participate in the repair of CNS injury resulting from irradiation. Our proposal is focused on analyzing the effects of space radiation on these critical populations of multipotent neural stem and precursor cells. We have developed the technology to isolate and grow these cells in culture and will use a variety of techniques to analyze how cells exposed to space radiation change. We will monitor the time course of oxidative stress in these cells and how that depends on the dose, the dose rate and type of radiation applied. We will determine how radiation-induced oxidative stress impacts cellular proliferation, signaling between cells, cell fate and survival. Collectively our studies using both rodent and human neural stem and precursor cells will help NASA understand the acute and longer-lasting sequelae associated with the exposure of the CNS to the space radiation environment. These studies will facilitate efforts to more appropriately establish estimates of risk for radiation exposure to the CNS in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from these studies will be critical in enhancing our knowledge base of the risks associated with exposure of the brain to the radiation fields in space.
Task Progress:	Scope of Work - General We have completed many of our objectives during the funding period, and have demonstrated the functional importance of oxidative stress in the irradiated CNS. Our overarching goal was aimed at determining if/how low dose and low dose rate proton and HZE charged particle irradiation leads to oxidative stress in neural precursor cells, and whether oxidative stress constitutes a biochemical mechanism responsible for disrupting cellular physiology and promoting cognitive dysfunction. Our past data has demonstrated that the radiation response of multipotent neural precursor cells to x-rays, protons, and HZE nuclei involves an acute and persistent oxidative stress that can last for months. The importance of neural precursor cells to the repair and remodeling of the irradiated CNS is underscored by our data showing that the radiation-induced depletion of these cells is linked to the inhibition of neurogenesis and the onset of cognitive impairment. We believe oxidative injury is critical in this context, and plays a contributory if not causal role in determining neural precursor cell function, survival and fate. Our data collected over the last few years has led to several recent publications, the highlights of which are presented below: 1. Liao, A.C., Craver, B.M., Tseng, B.P., Tran, K.K., Parihar, V.K., Acharya, M.M and Limoli, C.L. Mitochondrial-targeted human catalase affords neuroprotection from proton irradiation. Radiat. Res., 180:1-6 (2013). This manuscript highlights the importance of removing hydrogen peroxide from the irradiated brain. Mice genetically manipulated to overexpress human catalase in the mitochondria were found to exhibit higher basal levels of neurogenesis and after low dose exposure to 150 MeV protons. These data suggest that interventions targeted to reduce oxidative stress in the mitochondria are neuroprotective. 2. Tseng, B.P., Lan, M.L., Tran, K., Acharya, M.M., Giedzinski, E. and Limoli, C.L. Characterizing low dose and dose rate effects in rodent and human neural stem cells exposed to proton and gamma irradiation. Redox Biology, In Press (2013). This manuscript details a great deal of work characterizing the response of cultured neural stem cells from both rodents and humans. These data reveal a marked predisposition of neural stem cells to exhibit radiation-induced oxidative stress. The functional importance of radiation-induced oxidative stress in regulating adaptive responses was also shown, as treatment of cells with the antioxidant N-acetylcysteine eliminated the protective effect of prior exposure on cell survival following exposure to a subsequent higher dose of protons. Collectively, these data provide further evidence in support of the ides that space relevant fluences of charged particles elicit significant levels of oxidative stress that impact survival and proliferation. 3. Tseng, B., Giedzinski, E., Izadi, A., Suarez, T., Lan, M., Tran, K., Acharya. M.M., Nelson, G.A., Raber, J., Parihar, V and Limoli, C.L. Functional consequences of radiation-induced oxidative stress in cultured neural stem cells and the brain exposed to charged particle irradiation. Antioxid. Redox Signaling., In Press (2013). This manuscript details years of work characterizing the effect of HZE particle irradiation on cultured neural stem cells and the irradiated rodent brain. The paper documents the marked and persistent increases in oxidative stress found months after exposure to very low fluences of 600 MeV iron ions. Oxidative stress measurement made in vivo found that low dose exposure caused marked changes in the level and activity of several key antioxidant enzymes 2 weeks after exposure that returned to background 1month after irradiation. Importantly, exposure to 5 or 30 cGy of 600 MeV oxygen or titanium ions was found to elicit significant deficits in novel object recognition 12 weeks after expoure. These data link key changes in redox metabolism to functional changes in cognition, changes that have important implications for astronauts exposed to the space radiation environment. 4. Krasieva, T.B., Giedzinski, E., Tran, K., Lan, M.L., Limoli, C.L. and Tromberg, B.J. Probing the impact of gamma-irradiation on the metabolic state of neural stem and precursor cells using dual-wavelength intrinsic signal two-photon excited fluorescence. Journal of Innovative Optical Health Sciences. 4, Britton Chance Special Issue 3:289-300 (2011). This manuscript has detailed our assessment of the metabolic activity of neural stem cells after low dose irradiation using non-invasive 2-photon spectroscopy. 5. Acharya, M., Lan, M., Kan, V., Patel, N., Giedzinski, E., Tseng, B.P. and Limoli, C.L. Consequences of ionizing radiation-induced damage in human neural stem cells. Free Radic. Biol. Med. 49:1846-1855 (2010). This is a comprehensive study that has defined the radiobiological response and consequences of human neural stem cells exposed to low LET radiation.

	6. Globus, R.K., Caiozzo, V.J., Acharya, M., Fike, J.R. and Limoli, C.L. In Oxidative Stress in Applied Basic Research and Clinical Practice: Cancer, Redox regulation of stem cell compartments: The convergence of radiation-induced normal tissue damage and oxidative stress. (Eds.' D.R. Spitz, K.J. Dornfeld, D. Gius and K. Krishnan), Humana Press Inc. Totowa, NJ, pp.169-192 (2012). A review of the redox stress biology of irradiated neural stem cells and stem cells from other organ systems. 7. Work is being finalized to ascertain the impact of altering the microenvironmental redox state using EC-SOD deficient neural stem cells In summary, we have obtained significant data over the course of our funding period demonstrating that low dose charged particle irradiation can elicit oxidative stress in multipotent neural stem and precursor cells, derived from both rodents and humans. Our data shows conclusively that prior exposure elicits adaptive responses that are capable of attenuating oxidative stress and survival. Collectively our data suggest that radiation-induced oxidative stress constitutes a biochemical mechanism of functional significance to neural stem cell biology and cognition. We have been able to finalize a great deal of data during our No Cost Extension, which has led to 3 high profile papers in 2013. Additional remaining data will continue to be finalized for sections of various manuscripts that will integrate into ongoing NASA funded studies. We are grateful to NASA for the opportunity to conduct our research into the acute and chronic effects of charged particle irradiation on neural stem cells and the intact CNS, and believe that this funding has led to fundamental insights into space radiobiology. Findings from these studies should prove useful in efforts to more accurately estimate the uncertainties and risks associated with exposure of the CNS to the space radiation environment.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Liao AC, Craver BM, Tseng BP, Tran KK, Parihar VK, Acharya MM, Limoli CL. "Mitochondrial-targeted human catalase affords neuroprotection from proton irradiation." Radiation Research. 2013 Jul;180(1):1-6. http://dx.doi.org/10.1667/RR3339.1 ; PubMed PMID: 23672429 , Jul-2013
Articles in Peer-reviewed Journals	Tseng BP, Lan ML, Tran, KK, Acharya MM, Giedzinski E, Limoli CL. "Characterizing low dose and dose rate effects in rodent and human neural stem cells exposed to proton and gamma irradiation." Redox Biology. 2013;1(1):153-62. http://dx.doi.org/10.1016/j.redox.2013.01.008 , Jul-2013
Articles in Peer-reviewed Journals	Tseng B, Giedzinski E, Izadi A, Suarez T, Lan M, Tran K, Acharya M, Nelson G, Raber J, Parihar VK, Limoli C. "Functional consequences of radiation-induced oxidative stress in cultured neural stem cells and the brain exposed to charged particle irradiation." Antioxidants and Redox Signaling. 2014 Mar 20;20(9):1410-22. Epub 2013 Aug 12. http://dx.doi.org/10.1089/ars.2012.5134 ; PubMed PMID: 23802883 [note reported previously in July 2013 as Epub ahead of print] , Mar-2014
Articles in Peer-reviewed Journals	Baulch JE, Craver BM, Tran KK, Yu L, Chmielewski N, Allen BD, Limoli CL. "Persistent oxidative stress in human neural stem cells exposed to low fluences of charged particles." Redox Biol. 2015 Aug;5:24-32. Epub 2015 Mar 11. http://dx.doi.org/10.1016/j.redox.2015.03.001 ; PubMed PMID: 25800120; PubMed Central PMCID: PMC4371546 , Aug-2015
Journal/Magazine covers	Liao AC, Craver BM, Tseng BP, Tran KK, Parihar VK, Acharya MM, Limoli CL. "Cover in journal Radiation Research for article, 'Mitochondrial-targeted human catalase affords neuroprotection from proton irradiation.' " Radiation Research. 2013 Jul;180(1):1-6. http://dx.doi.org/10.1667/RR3339.1 , Jul-2013